ECON 5345 Homework 3 Report

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Question 1

All the four methods reflects high variation at (approximately) the same frequencies. This is consistent with the concept that the latter three are local averages of the periodogram.

The periodogram is not consistent in the sense that its variance does not vanishes. It is zigzagged since no smoothing is applied.

The AR spectrum is smoother among the four. This is due to its parametric nature.

Comparing the N-W kernel estimation with or without prewhitening, I find that the non-prewhitened one underestimates the variation at spiky frequencies.

The plots are in Figures 1 to 6.

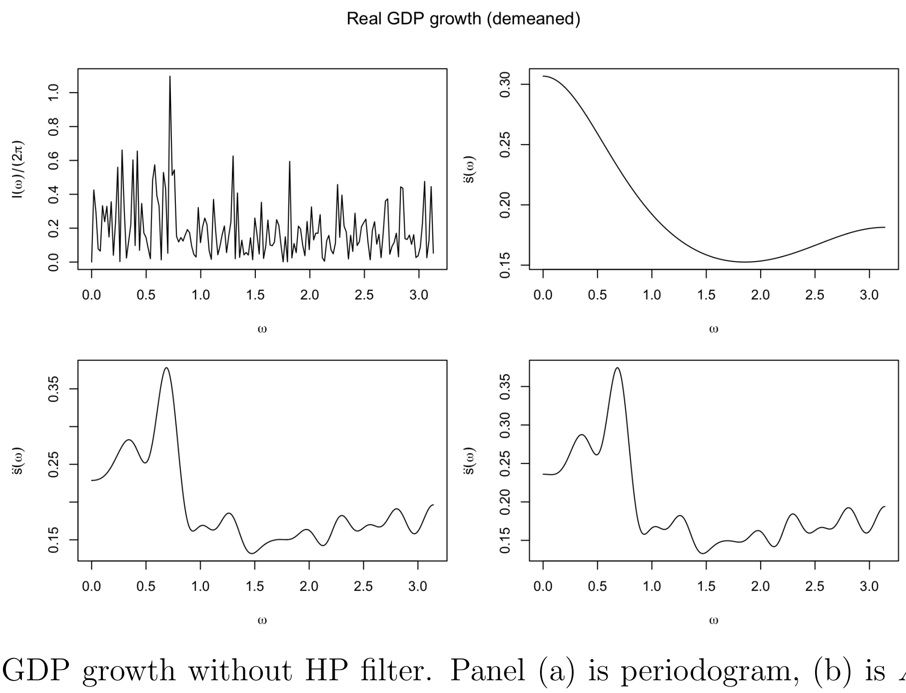

Figure 1: GDP growth without HP filter. Panel (a) is periodogram, (b) is $AR(2)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

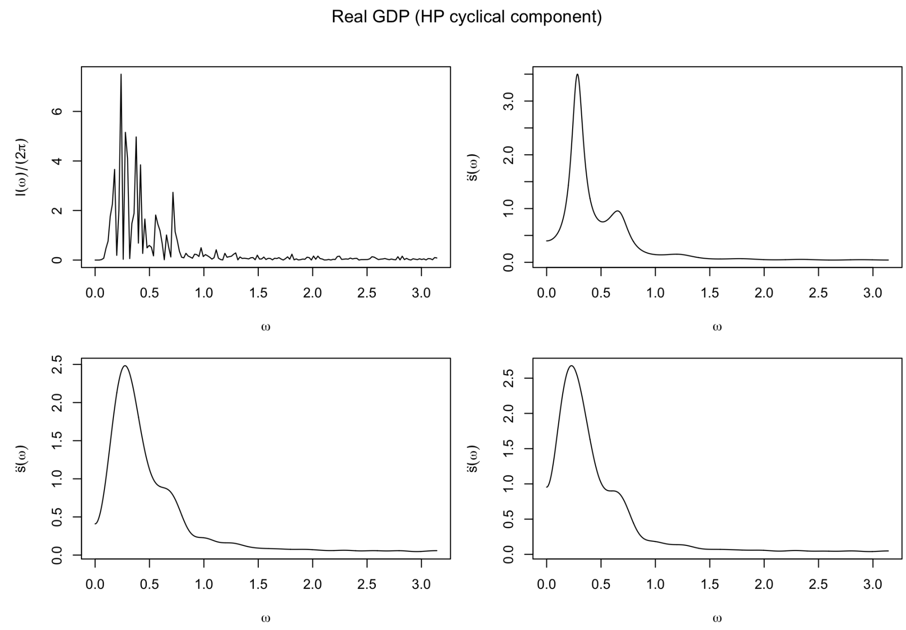

Figure 2: GDP growth with HP filter. Panel (a) is periodogram, (b) is $AR(12)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

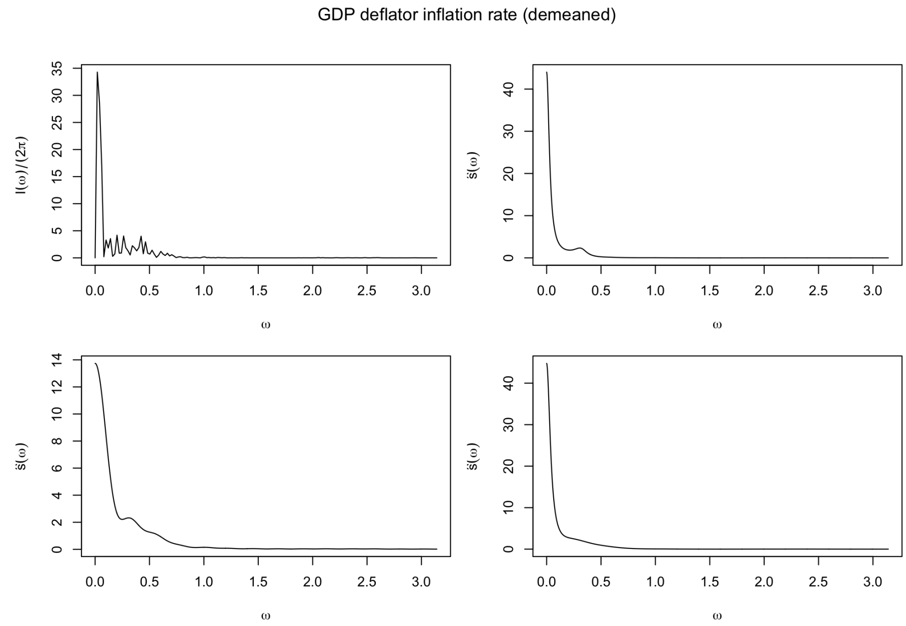

Figure 3: Inflation without HP filter. Panel (a) is periodogram, (b) is $AR(19)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

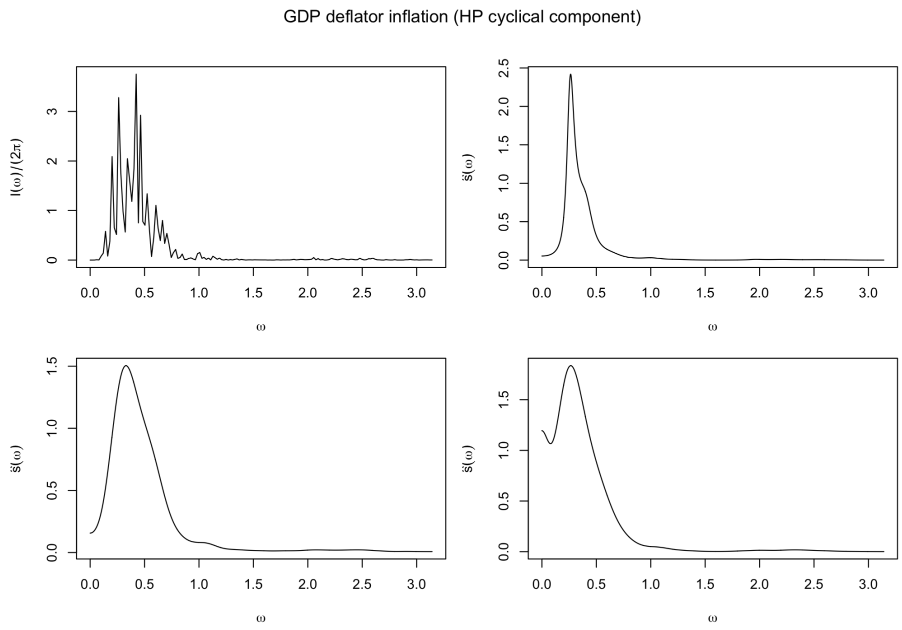

Figure 4: Inflation with HP filter. Panel (a) is periodogram, (b) is $AR(19)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

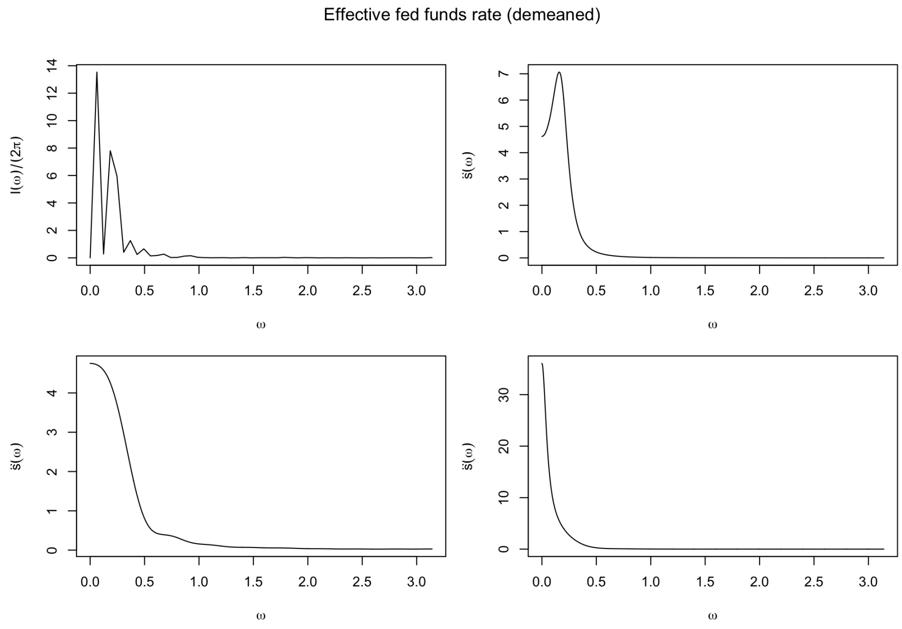

Figure 5: EFFR without HP filter. Panel (a) is periodogram, (b) is $AR(7)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

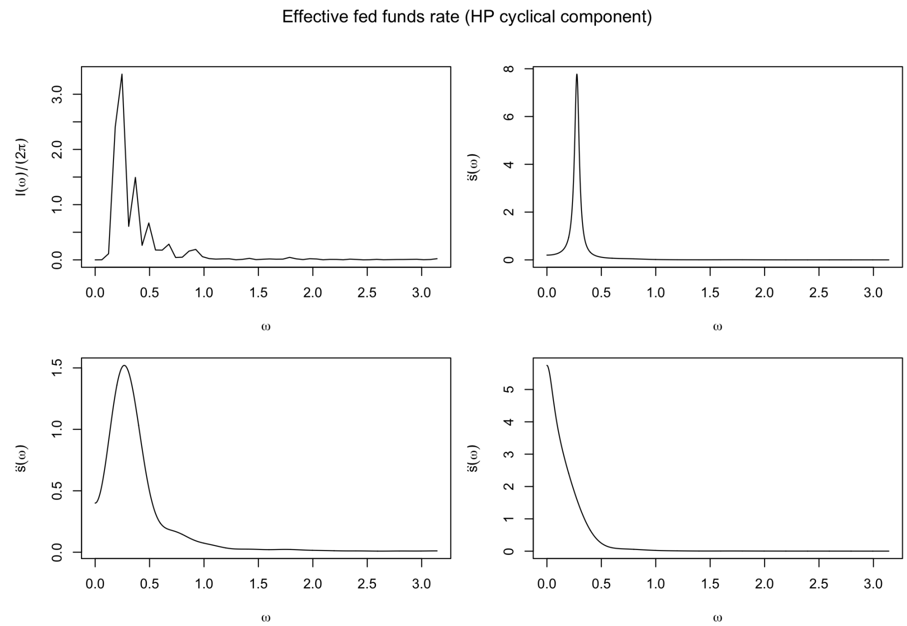

Figure 6: Effective FFR with HP filter. Panel (a) is periodogram, (b) is $AR(8)$ spectrum, (c) is N-W kernel spectrum, (d) is the prewhitened N-W.

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Question 2

Combine the two steps, we have

$$K(L) = \frac{1}{5}(L^{-5} - L^5)(L^{-2} + L^{-1} + 1 + L + L^2).$$

The gain is plotted in figure 7.

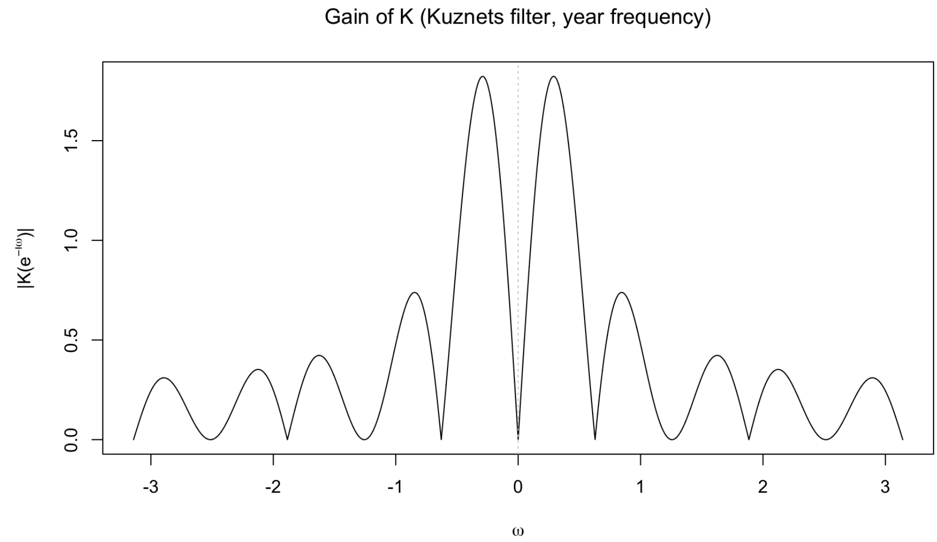

Figure 7: Gain of $K(L)$.

The last line of code gets the omega value that maximizes the gain (the peak of the curve), $\omega^* = 0.2901$. Note that since this is the year frequency transformation, this ω^* corresponding to a periodicity of $2\pi/0.2901 \approx 21.4$ years.

Expanding the expression of $K(L)$, we have

$$K(L) = \sum_{j=-7}^{7} \kappa_j L^j,$$

where

$$\kappa_j = \begin{cases} -\frac{1}{5} & \text{if } j = 3, 4, 5, 6, 7 \\ \frac{1}{5} & \text{if } j = -3, -4, -5, -6, -7 \\ 0 & \text{otherwise.} \end{cases}$$

It is essentially a band-pass filter. For a white noise process, it has flat spectral density. By applying this band-pass filter, the artificial swing will be generated.

Question 3

I assume $e_t \sim \mathcal{N}(0,1)$ and $u_t \sim \mathcal{U}(-\sqrt{3}, \sqrt{3})$ for data generation. I checked the sample moment $\mathbb{E}[e_t u_t] = 0.0025$, consistent with orthogonality assumption.

a. The following results are for HP-filter.

Statistic	Value
Min	-0.5874
1st Qu.	-0.1134
Median	0.0092
Mean (μ^{HP})	0.0075
3rd Qu.	0.1215
Max	0.5501
SD (σ^{HP})	0.1707
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 1\sigma^{HP})$	0.1550
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 2\sigma^{HP})$	0.0240
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 3\sigma^{HP})$	0.0005

Table 1: Summary statistics and upper-tail shares of simulated correlations for HP filter.

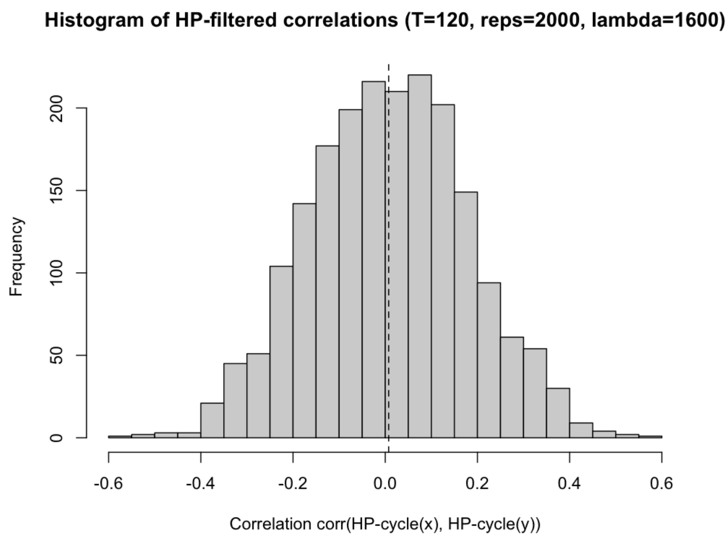

Figure 8: Histogram of HP-filtered correlations.

b. The following results are for Baxter-King filter.

Statistic	Value
Min	-0.6671
1st Qu.	-0.1432
Median	0.0135
Mean (μ^{BK})	0.0092
3rd Qu.	0.1619
Max	0.7416
SD (σ^{BK})	0.2171
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 1\sigma^{BK})$	0.1655
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 2\sigma^{BK})$	0.0225
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 3\sigma^{BK})$	0.0005

Table 2: Summary statistics and upper-tail shares of BK-filtered correlations (T=120, reps=2000, K=12, periods [6,32]).

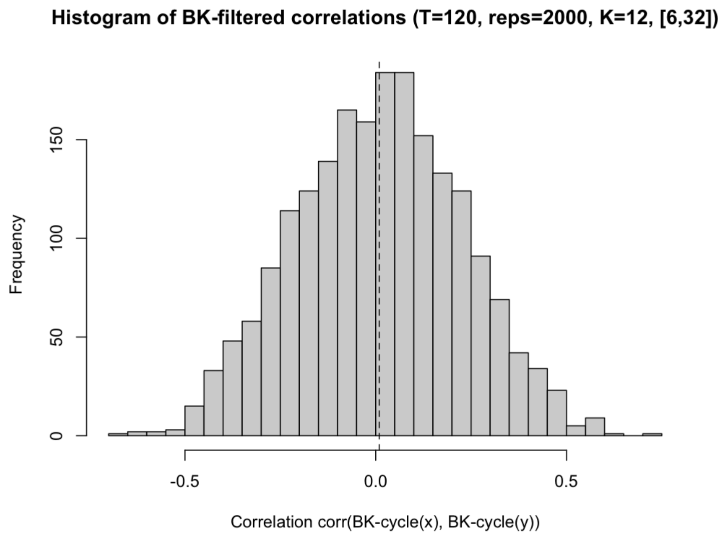

Figure 9: Histogram of BK-filtered correlations.

c. Discussion: In both cases, the mean correlation is close to zero, as expected under independence. However, the distribution of ρ_{xy} is dispersed indicated by the standard deviation values 0.1707 (HP) and 0.2171 (BK), and similar to a normal distribution, indicated by $\mathbb{P}(\rho_{xy} > \mu + z\sigma)$ values. For extreme values, correlations as large as 0.55 (HP) and 0.74 (BK) occur in the simulations.

Question 4

a. The gain of H^+ is plotted in Figure 10. The highest value is at around $|\omega| = 0.25 \in \left(\frac{2\pi}{32}, \frac{2\pi}{6}\right)$ which is in the range of the business cycle frequency. Therefore, $\{c_t\}$ will have more variation at the business cycle frequencies.

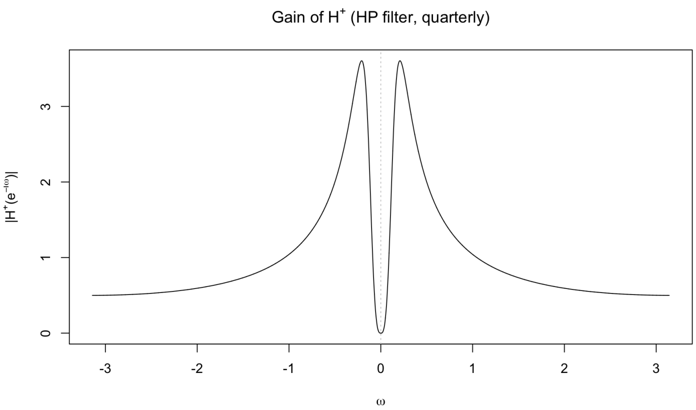

Figure 10: Gain of H^+

b. Since $c_t = H^+(L)y_t$, we have

$$s_c(\omega) = \left|H^+\left(e^-i\omega\right)\right|^2 s_y(\omega).$$

Since $y_t = x_t - x_{t-1} = e_t \sim WN(0, \sigma^2)$, we have $s_y(\omega) = \frac{\sigma^2}{2\pi}$. Therefore, the spectrum of $\{c_t\}$ is

$$s_c(\omega) = \frac{\sigma^2}{2\pi}\left|H^+\left(e^-i\omega\right)\right|^2$$

c. By definition, $s_c(\omega) = \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)e^{-i\omega j}$. Multiply both sides by $e^{i\omega k}$ for some integer k and integrate over $(-\pi, \pi]$:

$$\begin{aligned}
\int_{-\pi}^{\pi} s_c(\omega)e^{i\omega k}d\omega &= \int_{-\pi}^{\pi}\frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)e^{i\omega(k-j)}d\omega \\
&= \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)\int_{-\pi}^{\pi}e^{i\omega(k-j)}d\omega \qquad \text{by LDCT} \\
&= \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)\times 2\int_{0}^{\pi}\cos\left((k-j)\omega\right)d\omega \\
&= \gamma_c(k),
\end{aligned}$$

where the last line is by the periodicity of the $\cos(\cdot)$ function.

d. The table and figure below show the results of autocovariances.

k	$\gamma_c(k)$
0	1.668259408
1	1.203349899
2	0.807403465
3	0.477351924
4	0.208784655
5	-0.003618249
6	-0.165723093
7	-0.283644620
8	-0.363507014
9	-0.411257601
10	-0.432526914

Table 3: Autocovariances of $\{c_t\}$.

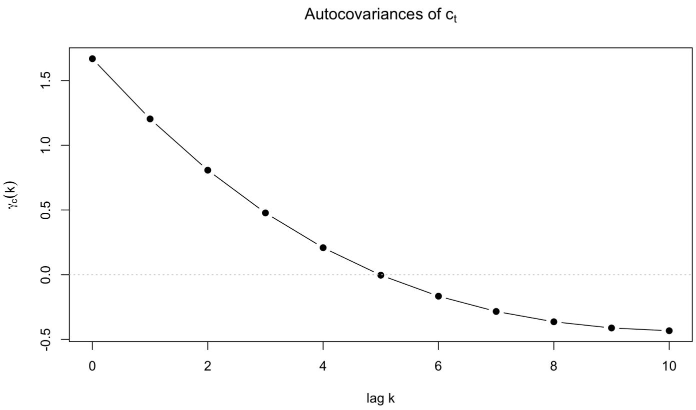

Figure 11: Autocovariances of c_t.